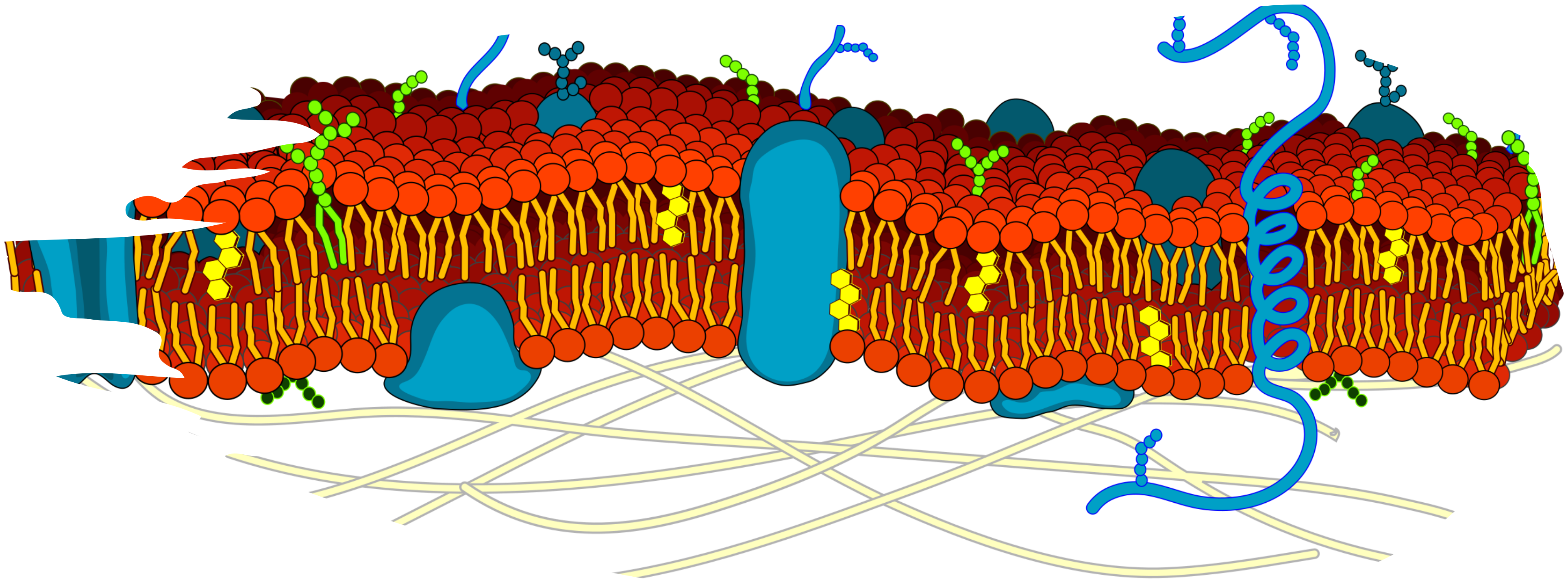


Cells I

BIOL2251

Chapter 3

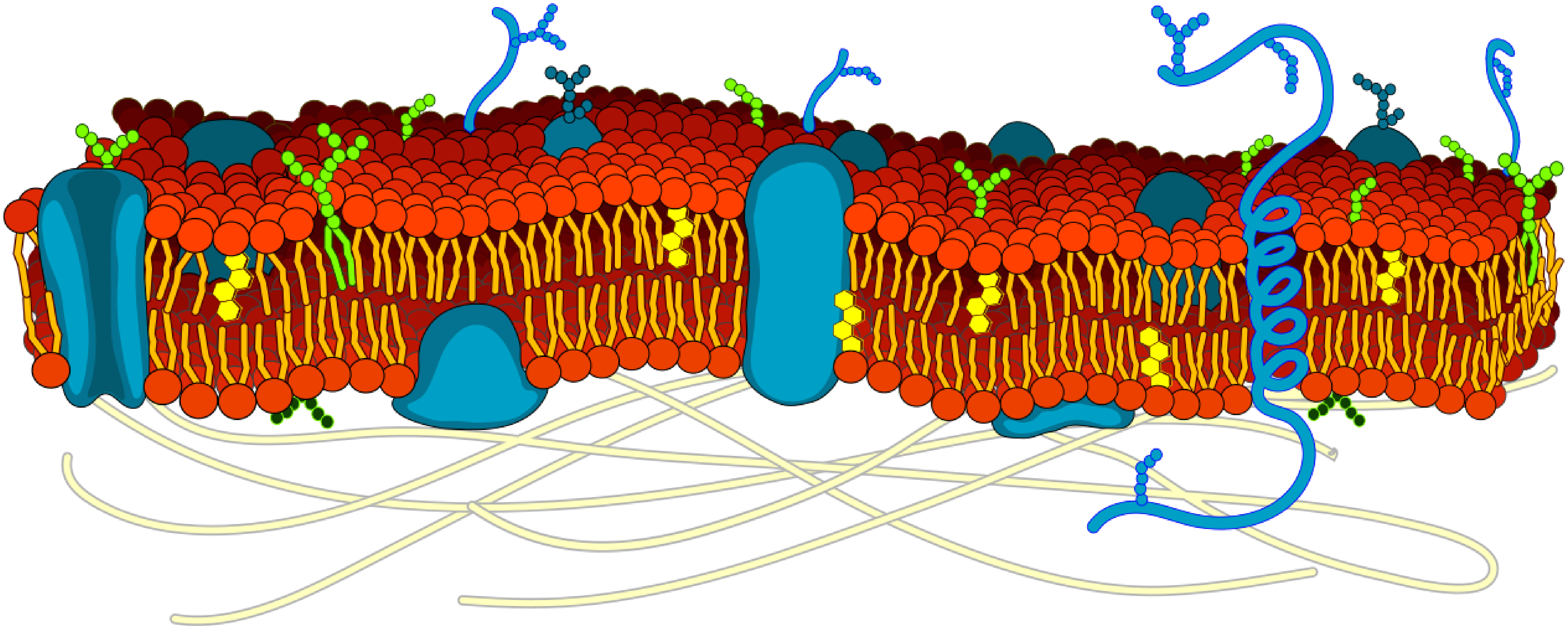
Sections 3.1-3.2, 3.5-3.7

Plasma Membrane Speed Build

Plasma Membrane Speed Build

- **Instructions**
  - Work in small groups (4-6 people)
  - I will project a list of membrane components
    - Phospholipids, cholesterol, peripheral proteins, integral proteins, glycoproteins, glycolipids
  - **You have 7 minutes to draw a simple diagram of a plasma membrane**
  - For each component, **recall a specific function it performs**
- **Bloom's taxonomy = Remember, Understand**

Plasma Membrane Speed Build



Plasma Membrane Speed Build

- **Phospholipid bilayer:** Creates selectively permeable barrier (hydrophobic core blocks water-soluble substances)
- **Cholesterol:** Stabilizes membrane fluidity; prevents membrane from becoming too fluid or too rigid
- **Integral proteins:** Span the membrane; function as channels, carriers, receptors, or enzymes
- **Peripheral proteins:** Attach to membrane surfaces; provide structural support, enzyme activity, or cell signaling
- **Glycoproteins:** Proteins with carbohydrate chains; cell recognition, immune response, receptor function
- **Glycolipids:** Lipids with carbohydrate chains; cell recognition and stability

Organelle Tournament

Organelle Tournament

- **Instructions**
  - I will present a **situation and two organelles competing for which is best**
  - **Vote for the organelle you think wins in the situation**
    - Votes cast by raising hands
  - After each match-up, we will discuss the reasoning
- **Bloom's taxonomy = Understand, Evaluate**

Organelle Tournament

1. **Protein synthesis: Ribosome vs. Golgi**
2. **Muscle cell abundance: Mitochondria vs. Peroxisomes**
3. **Detox specialist: Peroxisome vs. Lysosome**
4. **Digestive power: Lysosome vs. Proteasome**
5. **Surface area increase: Microvilli vs. Cilia**

Organelle Tournament

- **Protein synthesis: Ribosome vs. Golgi** - Ribosome (actually makes proteins; Golgi modifies afterward)
- **Muscle cell abundance: Mitochondria vs. Peroxisomes** - Mitochondria (muscle needs massive ATP for contraction)
- **Detox specialist: Peroxisome vs. Lysosome** - Peroxisome (breaks down toxins and fatty acids)
- **Digestive power: Lysosome vs. Proteasome** - Lysosome (digests larger materials; proteasomes handle tagged proteins)
- **Surface area increase: Microvilli vs. Cilia** - Microvilli (specifically for absorption surface area; cilia move substances)

Organelle Comparison Chart

Organelle Comparison Chart

- **Instructions**
  - You will be presented an empty chart
  - **Identify if the organelle is membranous or non-membranous and recall a key function**
  - Work on your own for 5 minutes, then pair with a partner to fill in gaps
  - We will clarify questions as a class
- **Bloom's taxonomy = Remember, Understand, Analyze**

Organelle	Membranous/Non-membranous	Key Function
Nucleus	Membranous (double)	Stores DNA; location of transcription (DNA to mRNA)
Nucleolus	Non-membranous	Produces ribosomal RNA; assembles ribosomal subunits
Ribosomes	Non-membranous	Location of translation (mRNA to protein); free or bound
Rough ER	Membranous	Protein synthesis and modification (secondary and tertiary structure); produces replacement membrane proteins
Smooth ER	Membranous	Synthesis of lipids, fatty acids, and carbohydrates; detoxification of drugs and toxins
Golgi Apparatus	Membranous	Protein/lipid modification and packaging
Mitochondria	Membranous (double)	ATP production (cellular respiration)
Lysosomes	Membranous	Digestion of cellular waste
Peroxisomes	Membranous	Break down fatty acids (produce H_2O_2 as a byproduct that is broken down into H_2O)
Proteasomes	Non-membranous	Digestion of proteins
Microvilli	Membranous extensions	Increase surface area for absorption
Cilia	Membranous extensions	Move materials across cell surface
Microfilaments	Non-membranous	Anchor cytoskeleton to integral proteins; determine consistency of cytosol
Intermediate Filaments	Non-membranous	Strength cell to maintain its shape; stabilize position of organelles; stabilize position of the cell relative to other cells
Microtubules	Non-membranous	Cell shape and movement; positioning of organelles; monorail system for motor proteins; distribute chromosomes during mitosis + meiosis
Centrosome	Non-membranous	Organizes microtubules
Centrioles	Non-membranous	Form spindle fibers during cell division

Cell Type Specialization

Cell Type Specialization

- **Instructions**
  - Work in small groups (4-6 people)
  - I will give you a specialized cell type
    - Muscle fiber, pancreatic cell, white blood cell, intestinal epithelial cell, neuron, sperm cell
  - **Discuss which 2-3 organelles would be most abundant in your cell type**
  - Be prepared to explain why
- **Bloom's taxonomy = Understand, Analyze, Apply**

Cell Type Specialization

- **Muscle cell:** Mitochondria (high ATP demand for contraction); smooth ER (calcium storage for contraction)
- **Pancreatic cell (makes digestive enzymes):** Rough ER (protein synthesis); Golgi apparatus (packaging enzymes for secretion); secretory vesicles
- **White blood cell:** Lysosomes (digest pathogens); Golgi (package enzymes)
- **Intestinal epithelial cell:** Microvilli (increase absorption surface area); mitochondria (active transport of nutrients requires ATP)
- **Neuron:** Rough ER and Golgi (makes neurotransmitters); mitochondria (axon transport and signaling require ATP); extensive cytoskeleton (maintains long processes)
- **Sperm cell:** Mitochondria (packed in midpiece for motility); flagellum with microtubules (movement)

Selective Permeability

Selective Permeability

- **Instructions**
  - Work in pairs
  - I will present **different molecules trying to pass through the plasma membrane**
    - Oxygen, glucose, Na ion, steroid hormone (testosterone), water, protein
  - **Predict whether the molecule can pass freely, needs help, or cannot directly pass the plasma membrane**
- **Bloom's taxonomy = Understand, Apply**

Selective Permeability

- **O_2 (oxygen gas):** EASILY crosses - small, nonpolar, lipid-soluble
- **Glucose:** NEEDS HELP - large, polar molecule requires carrier protein (facilitated diffusion)
- **Na^+ (sodium ion):** NEEDS HELP - charged particle requires channel protein
- **Steroid hormone (e.g., testosterone):** EASILY crosses - lipid-soluble, can pass through phospholipid bilayer
- **Water:** EASILY crosses (small, some movement through lipid bilayer) but faster through aquaporins (channel proteins)
- **Protein:** CANNOT cross directly - large, often charged; requires vesicular transport (endocytosis/exocytosis)

- **Key principle:** Lipid bilayer allows small, nonpolar (lipid-soluble) molecules to pass easily.

Osmosis Prediction Challenge

Osmosis Prediction Challenge

- **Instructions**
  - Work in small groups with mini-whiteboards
  - I will present 4 scenarios showing a cell in different solutions
  - In your groups:
    - **(1) Identify if the solution is isotonic, hypertonic, or hypotonic relative to the cell**
    - **(2) Draw and arrow showing net water movement**
    - **(3) Describe what happens to a red blood cell in that solution**
  - Hold up your board when ready to discuss
- **Bloom's taxonomy = Understand, Apply, Analyze**

Osmosis Prediction Challenge

- **Scenario 1:** Cell interior: 30 mm; External solution: 30 mm
- **Scenario 2:** Cell interior: 30 mm; External solution: 40 mm
- **Scenario 3:** Cell interior: 30 mm; External solution: 15 mm
- **Scenario 4:** Cell interior: 30 mm; External solution: 32 mm

Osmosis Prediction Challenge

Scenario 1: Cell interior: 30 mm; External solution: 30 mm

- **Tonicity:** Isotonic
- **Water movement:** No net movement (equilibrium)
- **RBC effect:** Normal shape; no change (this is normal saline)

Scenario 2: Cell interior: 30 mm; External solution: 40 mm

- **Tonicity:** Hypertonic (higher solute outside)
- **Water movement:** OUT of cell
- **RBC effect:** Crenation (shrivels/shrinks)

Osmosis Prediction Challenge

Scenario 3: Cell interior: 30 mm; External solution: 15 mm
- **Tonicity:** Hypotonic (lower solute outside)
- **Water movement:** INTO cell
- **RBC effect:** Hemolysis (swells and bursts)

Scenario 4: Cell interior: 30 mm; External solution: 32 mm
- **Tonicity:** Slightly hypertonic
- **Water movement:** OUT of cell (slowly)
- **RBC effect:** Slight crenation (mild shrinking)

Diffusion Factors

Diffusion Factors

- **Instructions**
  - Work in pairs or small groups (4-6 people)
  - I will provide **5 different scenarios comparing diffusion rates**
  - **Based on the scenario, predict which scenario/solution has a faster rate of diffusion**
- **Bloom's taxonomy = Understand, Analyze**

Diffusion Factors

Scenario A: Which diffuses faster?

- Solution 1: 100 molecules outside cell, 10 inside
- Solution 2: 60 molecules outside cell, 50 inside
- **Answer:** Solution 1
  - Steeper concentration gradient = faster diffusion

Scenario B: Which diffuses faster?

- Condition 1: Body temperature (37°C)
- Condition 2: Room temperature (22°C)
- **Answer:** Condition 1
  - Higher temperature = more kinetic energy = faster movement

Diffusion Factors

Scenario C: Which diffuses faster?

- Molecule A: Small oxygen molecule
- Molecule B: Large glucose molecule
- **Answer:** Molecule A
  - Smaller molecules diffuse faster

Scenario D: Which diffuses faster?

- Setup 1: 5 carrier proteins in membrane for glucose
- Setup 2: 20 carrier proteins in membrane for glucose
- **Answer:** Setup 2
  - More carriers = faster facilitated diffusion

Transport Mechanism Sorting

Transport Mechanism Sorting

- **Instructions**
  - Work in pairs
  - You will receive a table with transport mechanisms
  - **Complete the table by sorting transport mechanisms into passive/active and ATP requirement**
  - You will also **identify the type of material moved and provide an example of that transport mechanism in action**
- **Bloom's taxonomy = Remember, Understand, Analyze**

Transport Type	Passive/Active	ATP?	Material Moved	Example
Simple diffusion				
Osmosis				
Channel-mediated diffusion				
Facilitated diffusion				
Primary active transport				
Secondary active transport				
Exocytosis				
Receptor-mediated endocytosis				
Pinocytosis				
Phagocytosis				

Transport Type	Passive/Active	ATP?	Material Moved	Example
Simple diffusion	Passive	No	Small, nonpolar molecules	O_2, CO_2 crossing membrane
Osmosis	Passive	No	Water	Water moving into/out of cells
Channel-mediated diffusion	Passive	No	Small ions	Na^+, K^+ through ion channels
Facilitated diffusion	Passive	No	Large/polar molecules	Glucose via carrier protein
Primary active transport	Active	Yes (direct)	Ions against gradient	Na^+-K^+ pump
Secondary active transport	Active	Yes (indirect)	Various molecules	Glucose-sodium cotransport
Exocytosis	Active	Yes	Large molecules, waste out	Secretion of neurotransmitters
Receptor-mediated endocytosis	Active	Yes	Specific large molecules in	Cholesterol uptake via LDL receptors
Pinocytosis	Active	Yes	Fluid and dissolved solutes	"Cell drinking" - nonspecific fluid uptake
Phagocytosis	Active	Yes	Large particles/cells	White blood cell engulfing bacteria

Sodium-Potassium Pump: Step-by-Step

Sodium-Potassium Pump: Step-by-Step

- **Instructions**
  - Work on your own (to start)
  - **List the steps of the sodium-potassium pump in the correct order**
  - Compare your work with a partner
  - As a class, **we will clarify the steps and discuss why the sodium-potassium pump is vital**
- **Bloom’s taxonomy = Understand, Analyze**

Sodium-Potassium Pump: Step-by-Step

- 2 K^+ ions are released into the cytoplasm
- Phosphate group attaches to the pump, causing a shape change
- 2 K^+ ions from outside bind to the pump
- ATP binds and is hydrolyzed to ADP + phosphate
- The cycle repeats
- 3 Na^+ ions from the cytoplasm bind to the pump protein
- The pump returns to its original shape
- The phosphate group is released from the pump
- The shape change releases 3 Na^+ ions outside the cell

Sodium-Potassium Pump: Step-by-Step

1. 3 Na^+ ions from cytoplasm bind to pump protein
2. ATP binds and is hydrolyzed to ADP + phosphate
3. Phosphate group attaches to pump, causing shape change
4. Shape change releases 3 Na^+ ions outside the cell
5. 2 K^+ ions from outside bind to pump
6. Phosphate group is released from pump
7. Pump returns to original shape
8. 2 K^+ ions are released into cytoplasm
9. Cycle repeats

Sodium-Potassium Pump: Step-by-Step

Why it's important for resting membrane potential:

- Creates and maintains concentration gradients: high Na^+ outside, high K^+ inside
- Pumps 3 Na^+ out for every 2 K^+ in = net loss of positive charges inside
- Makes inside of cell more negative (contributes to -70mV resting potential)
- These gradients are essential for nerve impulses and muscle contraction
- Requires ATP (primary active transport) - uses ~30% of cell's energy at rest!

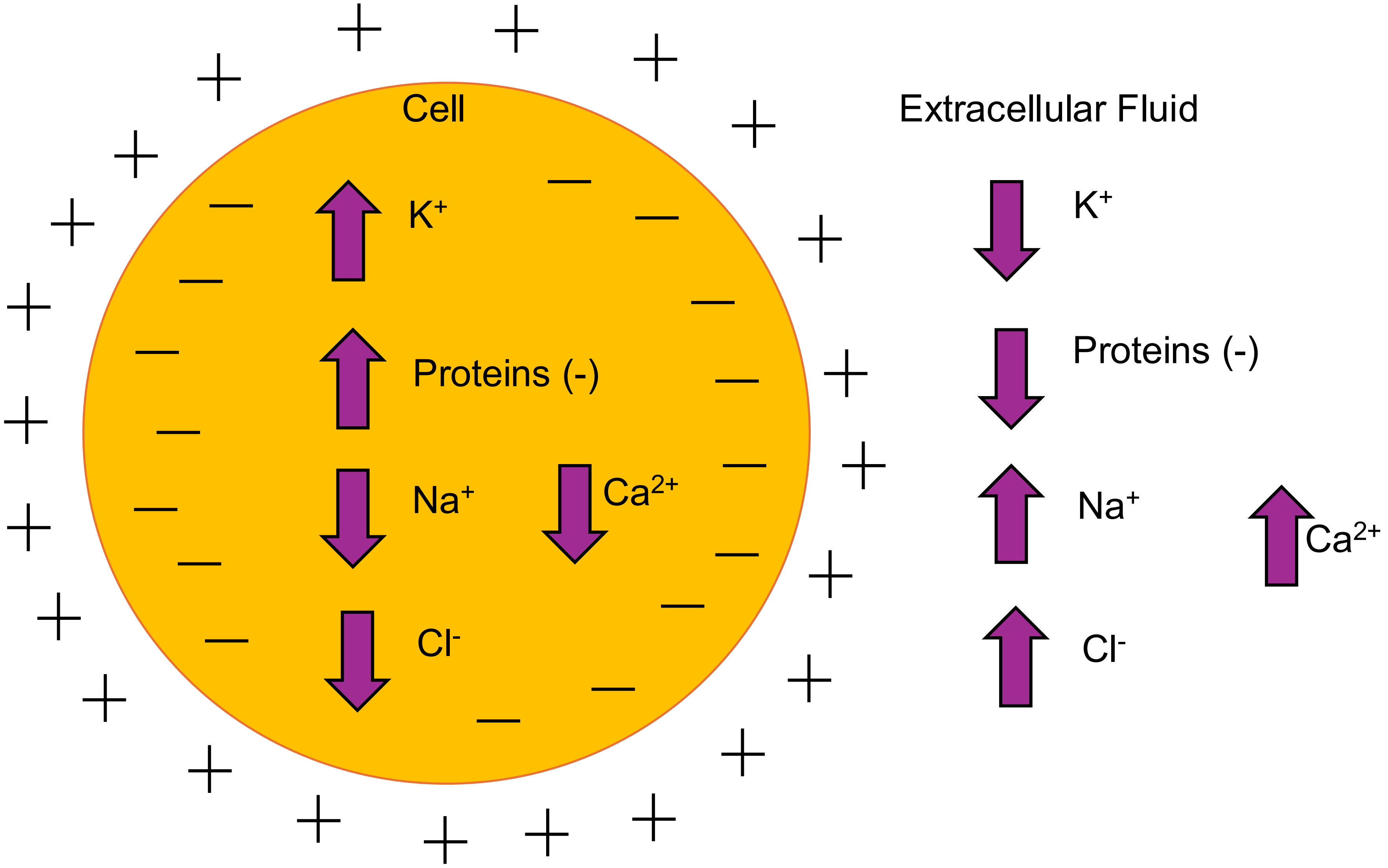
Membrane Potential Composition Comparison

Membrane Potential Composition Comparison

- **Instructions**
  - Work on your own
  - **Complete a comparison table showing the relative concentrations of key ions and overall charge**
    - Comparison is inside the cell (cytoplasm) and outside the cell (extracellular fluid, ECF)
  - Use this table and your knowledge of cell transport to answer the following: **why is the inside of the cell negative even though there is a lot K+ inside?**
- **Bloom's taxonomy = Remember, Understand, Apply**

Membrane Potential Composition Comparison

Component	Inside Cell (Cytoplasm)	Outside Cell (ECF)
Na^+		
K^+		
Cl^-		
Proteins (anions)		
Overall charge		


Cell
K^+
Proteins (-)
Na^+
Ca^{2+}
Cl^-
Extracellular Fluid
K^+
Proteins (-)
Na^+
Ca^{2+}
Cl^-

Membrane Potential Composition Comparison

Component	Inside Cell (Cytoplasm)	Outside Cell (ECF)
Na^+	LOW	HIGH
K^+	HIGH	LOW
Cl^-	LOW	HIGH
Proteins (anions)	HIGH	LOW
Overall charge	NEGATIVE (-70mV)	POSITIVE